

SMART INDIA HACKATHON 2026



- **Problem Statement ID – SIH26116**
- **Problem Statement Title-Urban Mixed-Use Design Challenge**
- **Theme-Vertical Synergy**
- **PS Category- Software**
- **Team ID-SIH26-A0H-T505**
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Urban Mixed-Use Design Challenge

EXPLANATION:

- The Ground and First Floors accommodate commercial spaces such as retail shops, cafés, offices, and co-working areas.
- The 2nd to 9th floors provide comfortable residential units with balconies, natural lighting, and ventilation.
- Separate commercial and residential entrances with a centralized lift and staircase core ensure efficient circulation, privacy, and safety.
- The building also has shaded façades and climate-responsive design strategies.

HOW IT WORKS:

- Reducing the need for separate developments maximises the available land.
- The basement solves urban parking requirements and includes **EV charging** for future mobility.
- Commercial spaces activate the street and provide economic activity, while upper floors create a live-work environment.
- Natural ventilation, daylight, shading, and greenery improve occupant comfort and reduce dependence on artificial systems.

INNOVATION AND UNIQUENESS :

- The key innovation is the concept of a “**Vertical Urban Hub**”, where mobility, commerce, work, and living are integrated into one connected building.
- The design uses a flexible commercial layout that can adapt to different businesses and future needs. EV-ready parking, green balconies, shaded façades, and a landscaped roof introduce sustainable features into a compact urban structure.

1. TECHNOLOGIES TO BE USED :

- Autodesk Revit – Architectural, structural, floor plans, elevations, sections, and 3D building modelling.
- Autodesk 3D Visualisation/Rendering – Present the proposed building through realistic 3D views.
- Dynamo for Revit – Optional visual programming for automation, repetitive modelling, schedules, and design optimisation.
- AutoCAD – Optional preparation and reference of 2D drawings and site information.
- Excel – Area statements, room schedules, parking calculations, and quantity documentation.

2. METHODOLOGY AND PROCESS FOR IMPLEMENTATION:

Problem Analysis → Site & Requirement Study → Concept Development → Site Planning → Space Zoning → Structural Grid → Revit Modelling → Sustainability and climate Integration → 3D Visualization → Documentation → Final Presentation

1. POTENTIAL CHALLENGES AND RISKS:

- Limited site area may create difficulties in accommodating parking, circulation, and open spaces. Conflicts may occur between commercial and residential circulation.
- High-rise construction requires careful structural and service coordination.
- Basement parking and EV charging require proper fire and electrical planning.
- Maintaining natural lighting and ventilation in a dense urban building can be challenging.

2. STRATEGIES FOR OVERCOMING THESE CHALLENGES:

- Use efficient space planning and a compact structural grid to maximize usable area.
- Provide separate entrances and circulation cores for commercial and residential users..
- Allocate dedicated areas for EV charging, electrical systems, fire safety, and building services.
- Use balconies, shaded windows, courtyards/voids, and façade fins to improve daylight and ventilation.
- Introduce variations in balconies, materials, screens, and greenery to create a dynamic façade.

- **POTENTIAL IMPACT ON THE TARGET AUDIENCE :**

Residents benefit from comfortable housing, secure access, natural light, ventilation, and convenient access to nearby commercial facilities. Businesses gain strategically located commercial spaces with high visibility and accessibility.

Visitors benefit from organized circulation, parking, and pedestrian-friendly spaces. The integrated development also creates a more connected and active urban environment by combining **living, working, shopping, and mobility** within one building.

- **BENEFITS OF THE SOLUTION :**

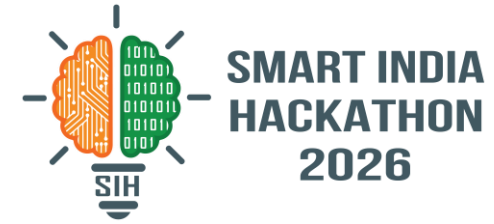
Economic Benefits:

Maximizes the value of limited urban land.
Generates income through commercial and residential spaces.
Supports local businesses and employment.

Social Benefits:

Creates a convenient live-work environment.
Provides safe and organized spaces for residents and visitors.
Encourages interaction through shared community areas.

RESEARCH AND REFERENCES



- **Mixed-Use Development & Sustainability — Nagpur, India**
[Research paper – Sustainability of Mixed-Use Neighbourhoods in Nagpur](#)
- **Vertical Mixed-Use Development**
[Research – Real Option and Vertical Mixed-Use Development](#)
- **Mixed-Use Development & Urban Sustainability**
[Research – Policy Coherence in Mixed-Use Development](#)
- **Mixed-Use Urban Form & Environmental Performance**
[Research – Mixed-Use and Open Urban Neighbourhoods](#)
- **National Building Code of India 2016**
[BIS – National Building Code of India 2016](#)
- **Autodesk Revit / BIM**
Autodesk also describes Revit workflows with Dynamo and sustainability analysis tools.
[Autodesk Revit for Architecture](#)